

Project Initialization and Planning Phase

Date	26 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Problem Statement:

Farmers often face difficulties in selecting the most suitable crop due to varying soil nutrient levels and changing environmental conditions. Traditional decision-making methods rely heavily on experience and intuition, which can result in poor crop selection, reduced productivity, inefficient use of fertilizers and water, and financial losses. There is a need for an intelligent, data-driven system that analyzes soil and climatic parameters to recommend the most suitable crop, helping farmers improve yield, optimize resource utilization, and support sustainable agricultural practices.

Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer planning the next cultivation season.	Select the most suitable crop for my farmland.	I am unsure whether my soil and weather conditions are appropriate.	I lack access to reliable, data-driven recommendations.	Uncertain about achieving a good harvest.

PS-2	A small-scale farmer.	Maximize crop yield while minimizing resource usage.	I rely on traditional farming practices and guesswork.	I have limited knowledge of soil nutrient analysis and crop suitability.	Concerned about low productivity and financial loss.
PS-3	An agricultural researcher.	Analyze crop suitability under different environmental conditions.	It is difficult to identify patterns from large agricultural datasets manually.	Multiple soil and climate factors influence crop growth simultaneously.	Challenged in making accurate, evidence-based recommendations.
PS-4	An agricultural policymaker.	Develop strategies to improve regional agricultural productivity.	I need reliable insights into crop-environment relationships.	Existing decisions often rely on limited or outdated information.	Motivated to make informed and sustainable policy decisions.